Matrix	Top mappings
$W_{OV}^{(43,25)}$	(Finnish, Finland) (Saskatchewan, Canadian) (Finland, Finnish) (Saskatchewan, Alberta) (Helsinki, Finnish) (Illinois, Chicago) (Chicago, Illinois) (Blackhawks, Chicago) (Australia, Australians) (NSW, Sydney) (Minneapolis, Minnesota) (Auckland, NZ) (Texas, Texans)
$W_{OV}^{(36,20)}$	(Japanese, Tokyo) (Koreans, Seoul) (Korean, Seoul) (Haiti, Haitian) (Korea, Seoul) (Mexican, Hispanic) (Spanish, Spanish) (Japanese, Japan) (Hawai, Honolulu) (Dublin, Irish) (Norwegian, Oslo) (Israelis, Jewish) (Vietnamese, Thai) (Israel, Hebrew)
$W_{OV}^{(31,9)}$	(oglu, Turkish) (Tsuk, Japanese) (Sven, Swedish) (Tsuk, Tokyo) (Yuan, Chinese) (stadt, Germany) (von, German) (Hiro, Japanese) (Sven, Norwegian) (stadt, Berlin) (Yuan, Beijing) (Sven, Danish) (oglu, Erdogan) (Fei, Chinese) (Samurai, Japanese)

Table 8: Example top mappings encoded in the parameters of attention heads that extract the attribute for many input queries.

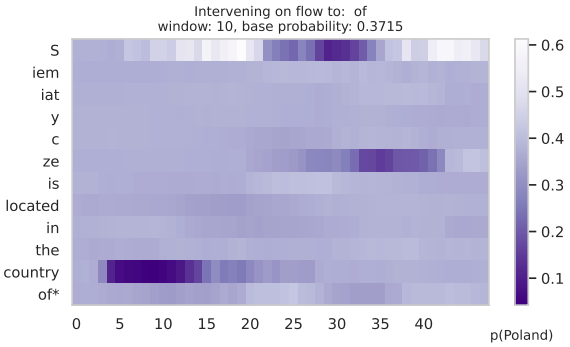

Figure 23: Example intervention on information flow to the last position of the input `Siemiatycze is located in the country of`, using a window size 10, in GPT-2.

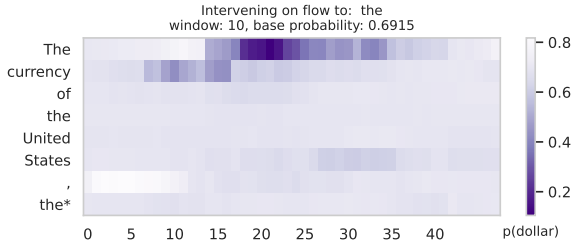

Figure 24: Example intervention on information flow to the last position of the input `The currency of the United States, the`, using a window size 10, in GPT-2.

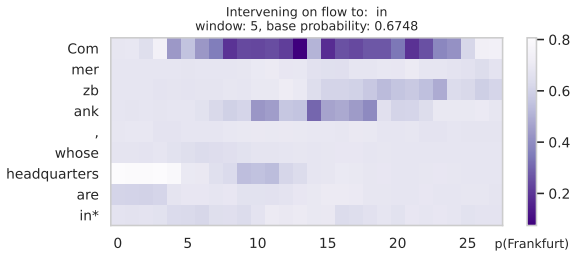

Figure 25: Example intervention on information flow to the last position of the input `Commerzbank, whose headquarters are in`, using a window size 5, in GPT-J.

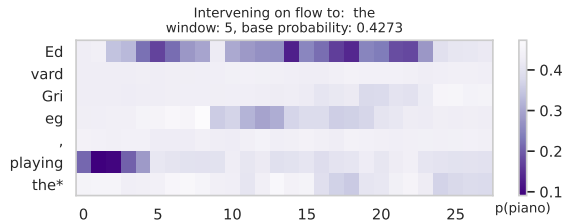

Figure 26: Example intervention on information flow to the last position of the input `Edvard Grieg, playing the`, using a window size 5, in GPT-J.

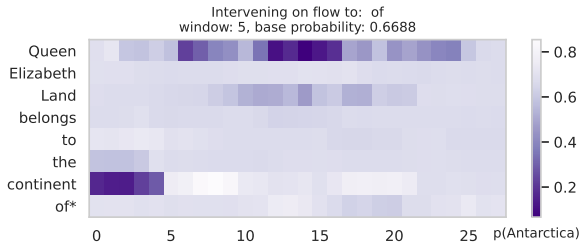

Figure 27: Example intervention on information flow to the last position of the input `Queen Elizabeth Land belongs to the continent of`, using a window size 5, in GPT-J.

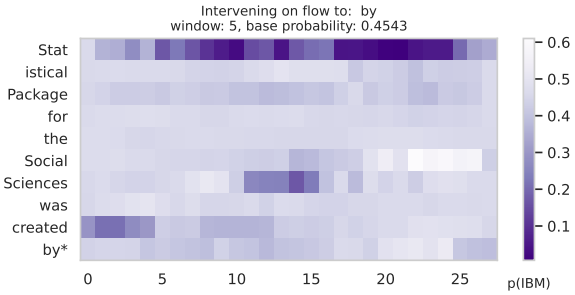

Figure 28: Example intervention on information flow to the last position of the input `Statistical Package for the Social Sciences was created by`, using a window size 5, in GPT-J.

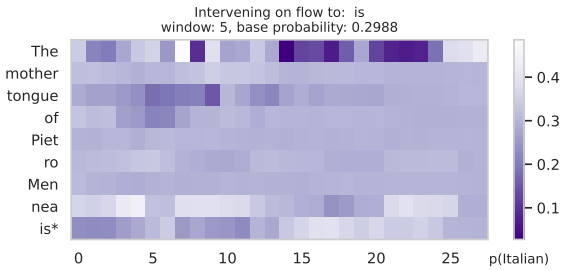

Figure 29: Example intervention on information flow to the last position of the input `The mother tongue of Pietro Mennea is`, using a window size 5, in GPT-J.